MUDIT KATRE

Full-Stack Developer | React.js, TypeScript, Python & FastAPI
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PROFILE

Final-year Computer Engineering student (B.Tech, 2027) who builds complete web applications, from React and Three.js interfaces to FastAPI and PostgreSQL backends. Smart India Hackathon 2024 national finalist, and team leader in SIH 2024, 2025 and 2026 and in L&T TECHgium 2026. Strong in leadership, teamwork and presentation skills, from leading hackathon teams and preparing idea presentations to serving as Secretary of StartBIT. Also a Data Analyst Intern.

EDUCATION

Bajaj Institute of Technology, Wardha
Bachelor of Technology (B.Tech) in Computer Engineering

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PHP, HTML5, CSS3
Frontend: React.js, Tailwind CSS, Three.js, React Three Fiber
Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy
Databases: PostgreSQL, MySQL, Database Design, SQL Query Optimization
AI/ML & Data Analytics: Pandas, NumPy, scikit-learn, XGBoost, PyTorch, SHAP, Exploratory Data Analysis (EDA), Power BI
DevOps & Testing: Git, GitHub, Docker, pytest, API Testing, CI/CD
AI Integration: LLM APIs, AI-powered application development

EXPERIENCE

Data Analyst Intern, Thiranex 2026
- Cleaned, structured, and analyzed real-world datasets using Python (Pandas, NumPy) and SQL for reporting
- Built Power BI dashboards and used EDA to find patterns and business insights

Cybersecurity Intern, NIIT Foundation 2026
- Completed a 4-week cybersecurity internship through AI Prime under the Cyber Suraksha Plus Program

PROJECTS

HomeGen: AI House Planner & 3D Visualization 2026
- Building a planner that creates house layouts within user-given dimensions, with interactive 2D and 3D views
- Adding material and construction cost estimates based on the user's requirements

Technologies: React.js, TypeScript, Tailwind CSS, React Three Fiber, Python, FastAPI, PostgreSQL, Shapely, Docker

VarshaMitra: Monsoon Rainfall Forecast Correction 2026
- Designing a system for NCMRWF (SIH 2026) that corrects monsoon rainfall forecast errors for each weather pattern
- Developing the pipeline in five modules: weather pattern classifier, error correction, heavy rainfall probability, district-level dashboard and daily accuracy checks

Technologies: Python, PyTorch (U-Net), XGBoost, scikit-learn, GeoPandas, SHAP, Streamlit, FastAPI, PostgreSQL, Docker

AICTE Sentinel: Approval Workflow & Compliance Monitoring 2026
- Designed a system for AICTE's Regulation Bureau (SIH 2026) to automate approvals, track compliance and send alerts
- Planned an 8-stage approval workflow with security first, using PostgreSQL for records and MinIO for document storage
- Prepared the technical approach, system flow diagram (draw.io) and idea presentation for submission

Technologies: React.js, FastAPI, Python, PostgreSQL, MinIO, REST APIs, Docker, draw.io

Verified Domestic Helper Management System 2024
- Built a web platform that connects households with verified drivers, cooks and maids, to make hiring safer
- Made the helper verification process and the service request flow

Technologies: HTML, CSS, PHP

LEADERSHIP EXPERIENCE

Team Leader, Smart India Hackathon (SIH) 2024 – 2026
- Led the team in SIH 2024, 2025 and 2026. The 2024 team reached the national final with a real-time monitoring system for Fire Department applications (inspections, follow-ups, NOC issuance); I built its dashboard frontend

Team Leader, L&T TECHgium Hackathon (10th Edition) 2026
- Led the team and prepared the Stage 1 abstract submission

Secretary, StartBIT (College Entrepreneurship & Startup Forum) 2023 – 2024
- Kept documentation for forum activities, including event reports and Minutes of Meeting (MOM)
- Maintained financial records of funds used across forum events

CERTIFICATIONS

- Foundations of Digital Marketing and E-commerce, Coursera, 2026
- Building a Startup, course by Ankur Warikoo